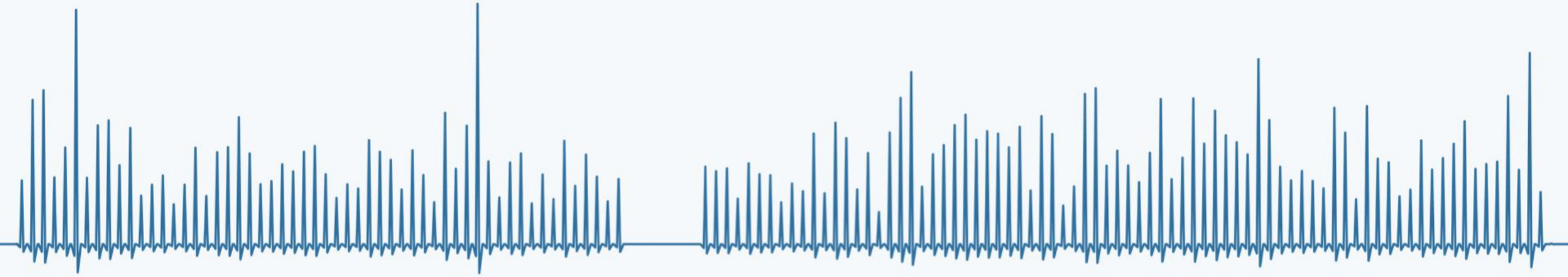
Noéa

Reading water as rhythm, memory and mood

Affective computing for water systems · Meeting of Waters, Geneva



Annual peak daily rainfall, Hong Kong, 1884–2025, drawn as a pulse. The flat stretch is 1940–46, missing from the record.

Data can show that a river is changing. It rarely makes anyone feel it.

Detect

Stations now measure rainfall, rivers and glaciers at scale.

The missing layer

Turning measurement into meaning people can experience.

Respond

Decisions stall between knowing and acting.

Noéa works in the middle.

It reads a record the way a cardiologist reads an ECG

A long record

Daily rainfall or river discharge, covering decades to centuries.

Diagnostics

Peaks and the gaps between them, return periods, flow duration, recession, the seasonal cycle.

Two affective values

Valence: pleasant ↔ distressing.

Arousal: calm ↔ activated.

Plus four physical readouts: size, glow, texture, tempo.

XR and sound

Designed to feed immersive and sonic experiences. The output format is still provisional.

Two layers, kept deliberately apart

Science layer

OBJECTIVE

Hong Kong rainfall, 1884–2025: annual peaks, the gaps between them, return periods.

Rhine and alpine rivers, 8 stations: flow duration, recession, the seasonal cycle.

Measures what happened. The measurements are never adjusted for artistic reasons.

Curatorial layer

DESIGNED

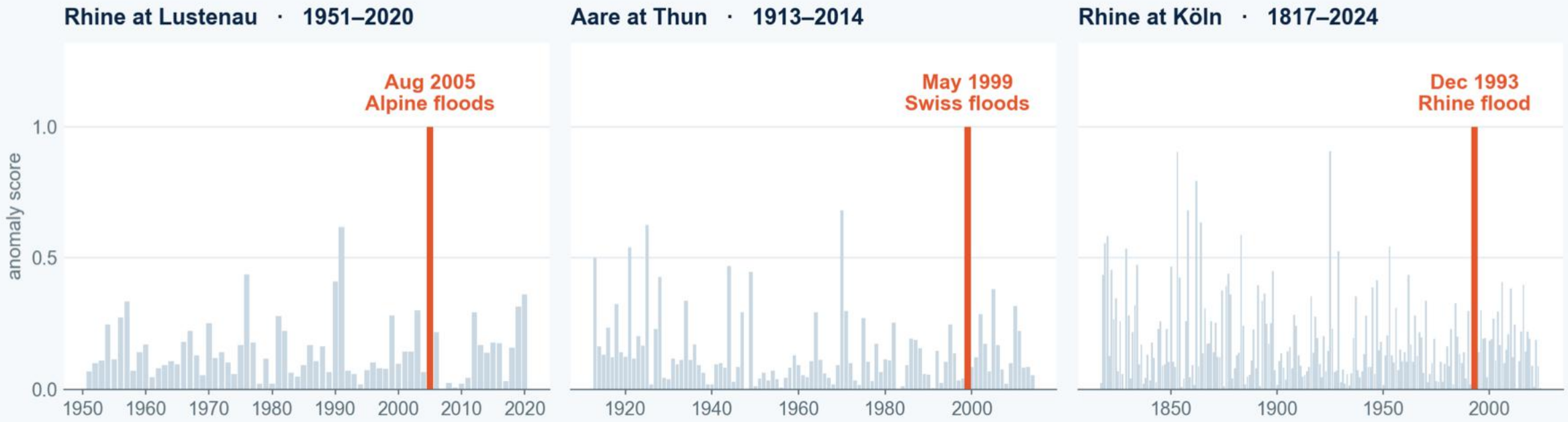
Decides what a drought or a flood should feel like: how alarming, how heavy.

Kept separate from the measurements, so the feeling can be redesigned without changing any of the science.

Its output: valence and arousal.

Machine learning finds unusual years

Its one job so far: for each record, an unsupervised model learns what a typical year looks like, then scores how badly each year fails to fit.



The highlighted bar is each station's highest-scoring year. The model was never told these events existed.

Method: PCA reconstruction error, fit separately per station on its own years. Events: Aug 2005 Alpine floods (22–23 Aug), May 1999 Swiss floods (Thun, 15 May), Christmas 1993 Rhine flood (Köln, 24 Dec).

Built to travel

Each record fits its own model on its own history. Another rainfall or river record can go through the same pipeline.

Hong Kong rainfall

HONG KONG OBSERVATORY

Daily, 1884–2025

One model, fit on 134 years

Rhine and alpine rivers

GRDC · FOEN

Daily, 8 stations

One model per station, 62–208 years each

The next record

NOT YET RUN

Another long rainfall or river series.

Tested on two records so far.

What it doesn't do yet

1 **It misses droughts.**
The model flags unusual combinations of features, not extremes along the wet–dry axis.

2 **The step from anomaly to emotion is designed by hand.**
It is not yet learned from people.

3 **Nobody has tested whether viewers feel what it intends.**

Next: learn the mapping from human responses, and test it with the people who make water decisions.